

HPE Digital Twin Project - Implementation Plan

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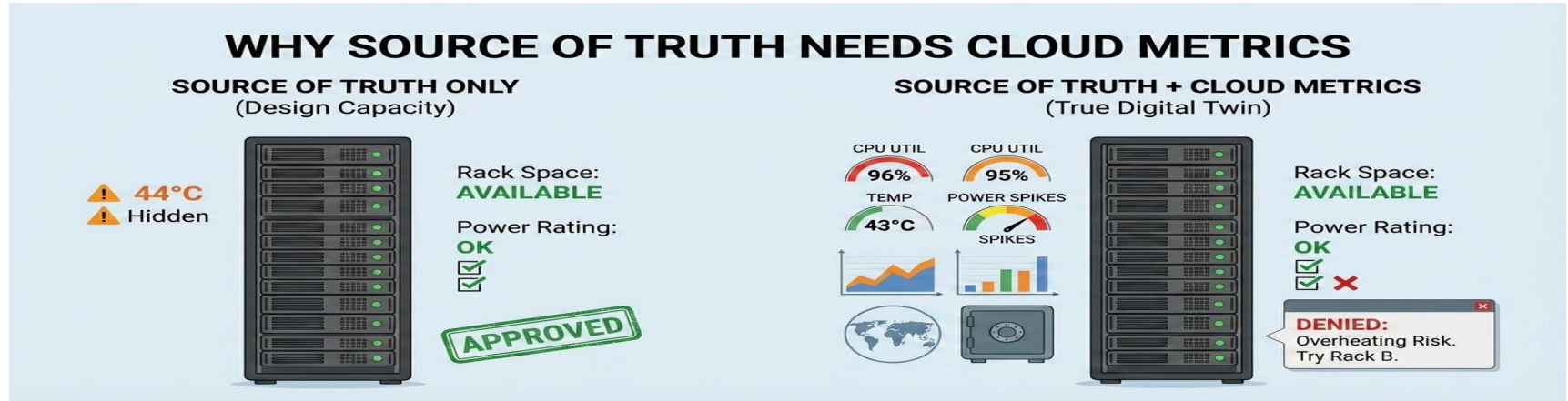
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Why Source of Truth needs Cloud Metrics



The Problem: Blind to Reality

SoT checks static limits (space, power) but ignores live data (CPU load, heat).

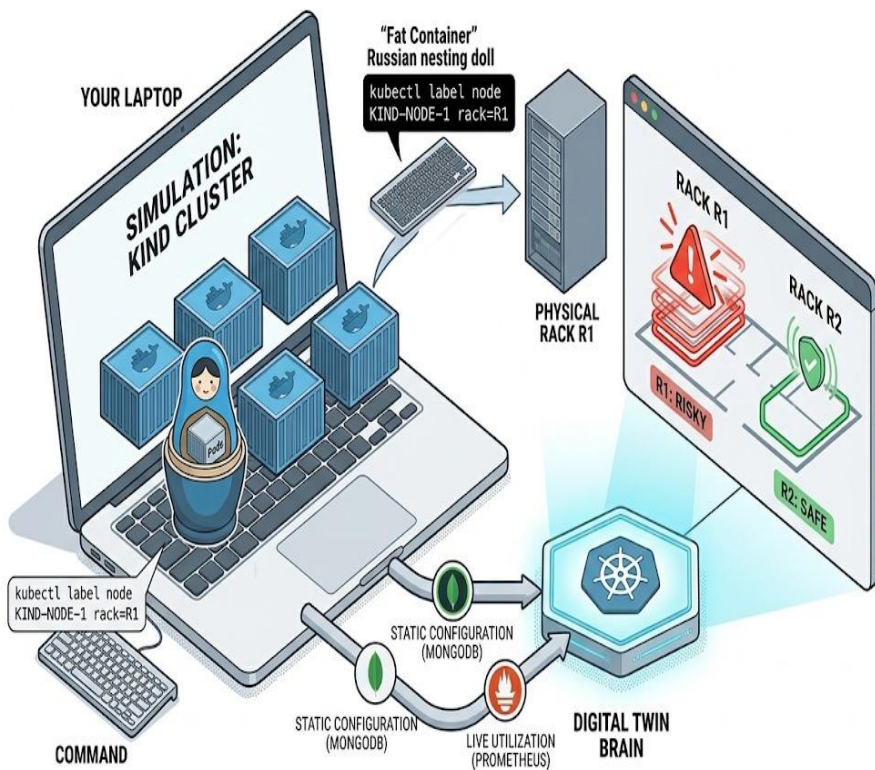
It can easily approve a deployment that physically overheats a server rack.

The Solution: The True Digital Twin

Merges static config rules with live cloud metrics.

Proactively detects risks before deployment and recommends the safest location (e.g., "Try Rack B").

Collecting metrics and simulating with Kind Cluster



1. **"Fake Servers":** Instead of simple apps, `kind` containers run full operating systems to act exactly like physical servers.

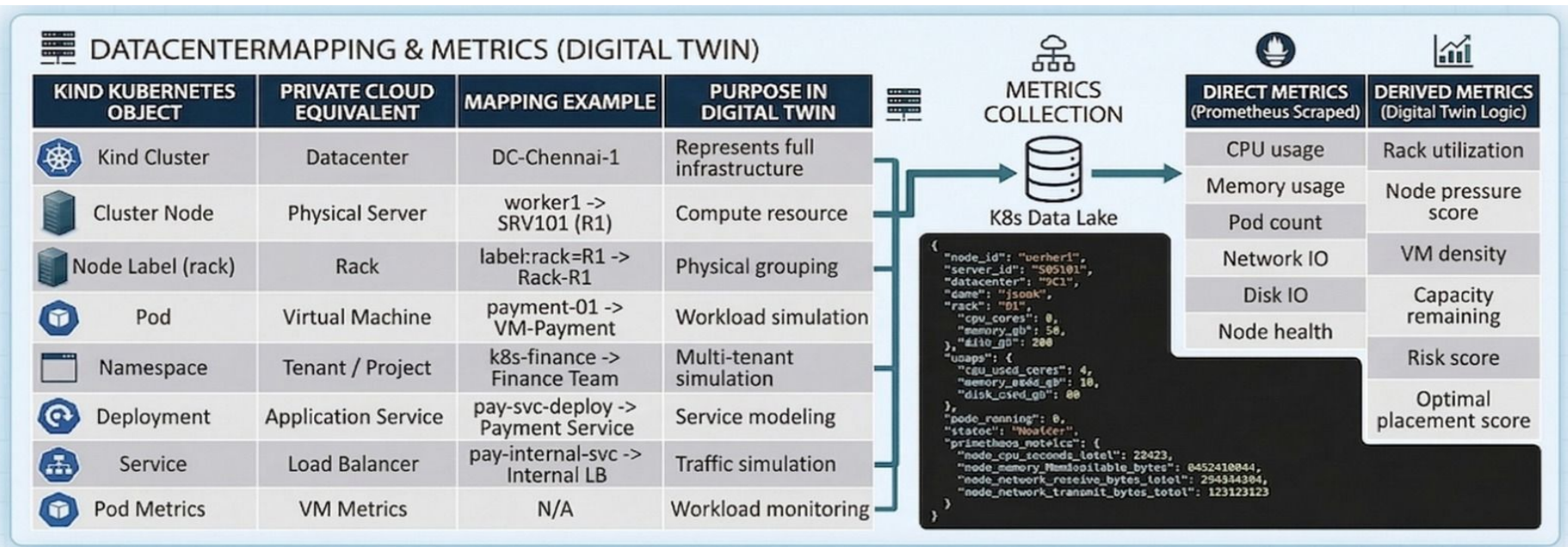
2. **Virtual to Physical:** We use Kubernetes labels (e.g., `rack=R1`) to group these virtual nodes into physical "Racks."

3. **Dual Data Streams:**

- **MongoDB (The Blueprint):** Provides static hardware limits (e.g., max power).
- **Prometheus (The Pulse):** Provides live runtime metrics (e.g., CPU, heat).

4. **The Digital Twin:** It fuses the static blueprint with live reality, creating a risk-free testing ground right on your laptop.

Collecting metrics and simulating with Kind Cluster



Kind cluster enables private cloud simulation by mapping Kubernetes nodes to physical servers and pods to virtual machines, allowing realistic infrastructure modeling without requiring actual hardware.

Unlike static simulations, Kind uses the real Kubernetes scheduler and Prometheus metrics, enabling realistic workload placement, scaling behavior, and infrastructure monitoring similar to a private cloud

Synthetic Data Generation Pipeline



1. Real Data Collection (Prometheus Baseline)

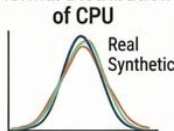
- CPU usage distribution
- Memory usage patterns
- Network traffic trends
- Node workload behavior

Node	CPU	[Memory]	Pods
worker1	[65%]	[60%]	[8]
worker2	[72%]	[68%]	[10]
worker3	[58%]	[55%]	[6]

Baseline used for synthetic generation



Normal Distribution of CPU



Statistical Similarity



2. Pattern Learning & Statistical Extraction

- Mean CPU usage
- Standard deviation
- Workload categories
- Time-based trends

Example: Mean CPU = 69%, Std Deviation = 8%

Synthetic Range: 60% – 78%

Pattern Driven Synthetic Generator

Synthetic Metric = Real Mean + Profile Factor + Time Pattern + Small Variation

Example: CPU = 69 + profile + time + random(-5,5)

Metric Relationships:

CPU ↑ → Temperature ↑ → Power ↑

Temperature: Temp = 28 + CPU × 0.15

Power: Power = Base + CPU factor



3. Synthetic Infrastructure Output

- Server behavior profiles applied
- Metric correlations maintained
- Time patterns simulated
- Synthetic infrastructure generated

Server Type	CPU	Memory	Network
[Compute]	High	Medium	Low
[Database]	Medium	High	Low
[Web]	Medium	Medium	High

Generated Synthetic Infrastructure Example

Server	CPU	[Memory]	[Temp]	Status
[SRV101]	[71%]	[62%]	[39°C]	Healthy
[SRV102]	[66%]	[58%]	[37°C]	Healthy
[SRV103]	[80%]	[75%]	[42°C]	Warning

Synthetic servers expand infrastructure scale (SRV101-SRV200)

Synthetic Data Follows Real Distribution

Key Benefits of Synthetic Data for Digital Twin



Simulates large datacenter scale



Tests failure scenarios



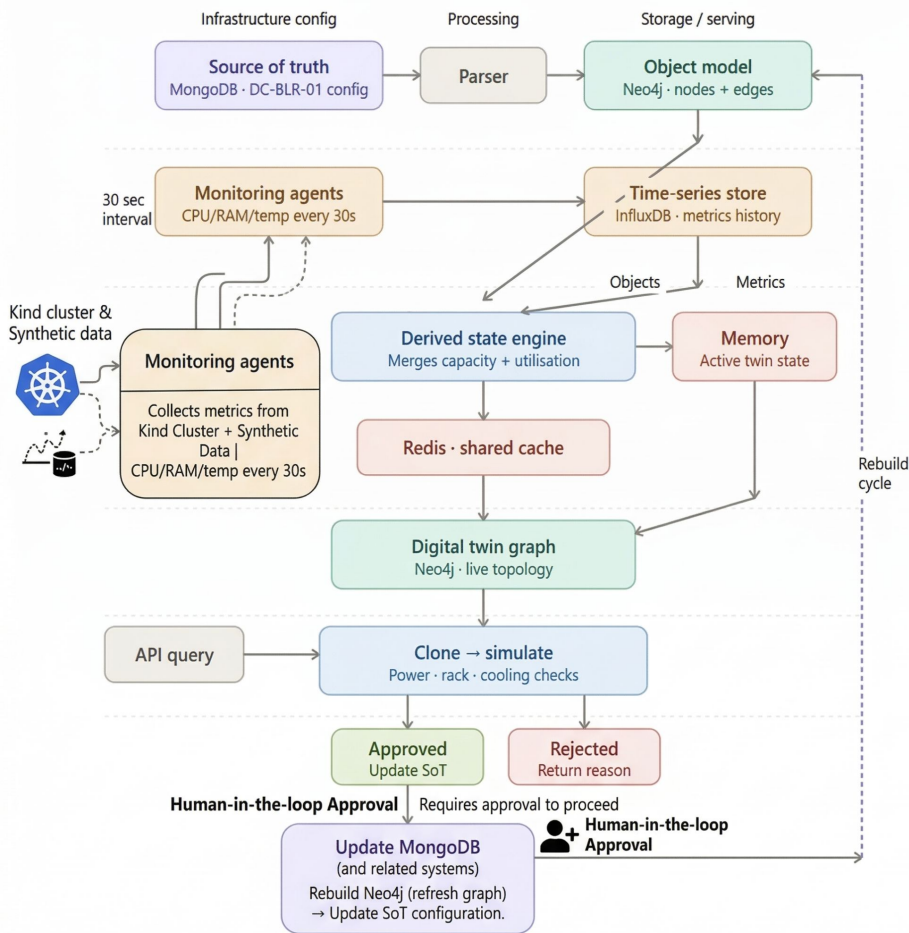
Enables placement testing



Models infrastructure growth

Synthetic infrastructure metrics are generated using statistical patterns derived from real Kubernetes telemetry to ensure realistic simulation.

System flow and Architecture



1. Dual Data Ingestion

- **Static Blueprint:** Source of Truth is stored in MongoDB as raw infrastructure configuration.
- This configuration is parsed into infrastructure objects and modeled as relationships inside Neo4j to build the Digital Twin topology.
- **Live Telemetry:** Agents pull current cloud metrics from the **kind** cluster every 30s into InfluxDB.

2. The Derived State Engine

- This central engine merges static capacity with live utilization.
- It caches the active state in Redis and builds a live, unified topology graph in Neo4j.

3. Isolated Simulation (Clone & Test)

- When an API query requests a change, the engine *clones* the live graph.
- It runs isolated "what-if" checks (Power, Rack, Cooling) without affecting the real system.

4. Human-in-the-Loop Execution

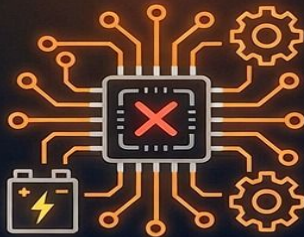
- The simulation outputs a definitive **Approved** or **Rejected** (with reasons).
- Approvals require a final human sign-off before the system safely updates the MongoDB Source of Truth.

THE FOUR STAGES OF PRIVATE CLOUD DIGITAL TWIN INTELLIGENCE



1. REACTIVE MONITORING (Reflex)

- Immediate Threshold Breaches. (e.g., CPU > 95%).
- Stop-the-Bleeding Responses.
- **Tech:** Prometheus Alertmanager. (deterministic rules).



2. DETERMINISTIC SIMULATION (Blueprint)

- “What-If” Topological Testing.
- Validate Physical Constraints. (PDU -> Switch -> Server).
- **Tech:** Neo4j Cypher & Sandbox Cloning. (probabilistic logic).



3. PREDICTIVE ANOMALIES (Intuition)

- Identify Hidden “Drifts” & Weak Signals. (e.g., Thermal Drift).
- Pattern Recognition Over Time.
- **Tech:** AI (Isolation Forest, LSTM). (pattern analysis).



4. PROACTIVE OPTIMIZATION (Advisor)

- Prescribe Optimal Future State.
- Multi-Objective Decision Making. (Power, Cool, Load).
- **Tech:** Prescriptive AI & Reinforcement Learning (multi-variable optimization).

From Immediate Alerts to Optimized Decision Support.

FOUR STAGES OF PRIVATE CLOUD DIGITAL TWIN INTELLIGENCE

Stage	Logic Type	Technical Driver	Why this is the “Apt” Approach?
1. Reactive	Deterministic	Prometheus Alertmanager	● Speed & Safety: Hardcoded thresholds (e.g., Temp > 90°C) provide an instant, 100% reliable reflex. AI is too slow and risky for “firefighting.”
2. Simulation	Topological	Neo4j + Source of Truth	● Physical Certainty: If a PDU is cut, the power loss is a mathematical fact, not a guess. Graph checks provide a transparent, audit-ready “Blueprint” of the blast radius.
3. Predictive	Probabilistic	AI (Isolation Forest / LSTM)	● Hidden Pattern Discovery: AI identifies “Drift” (e.g. rising heat vs. idle CPU) by comparing History Trends to the Baseline. Humans cannot write if then rules for 500+ shifting variables.
4. Proactive	Prescriptive	AI (Reinforcement Learning)	● Multi-Variable Optimization: Finding the “Best” migration path requires balancing power, cooling and latency. AI acts as the Advisor to solve trade-offs that exceed human calculation.

Unique value proposition

Key Innovations:

1 Live Digital Twin using Config + Cloud Telemetry

system combines **MongoDB config + Live cloud metrics** (Kind + Prometheus + InfluxDB) to **dynamically generate a real-time Digital Twin**

- Reflects actual runtime utilization
- Continuously updated twin (every 30 sec)
- Dependency-aware infrastructure modeling
- Supports real operational decision making

3 Multi-Database Architecture



- MongoDB for Infrastructure config
- Neo4j for Relationship topology
- InfluxDB for Time-series metrics
- Redis for Fast derived state access
- Memory for Active twin state

4 High Performance Execution



Derived infrastructure state stored in Memory + Redis cache

- Fast simulation execution
- Low latency decisions
- Real-time query response

2 Intelligent Recommendation Engine

Provides failure reason explanation, constraint violation details, alternative placement suggestions, resource optimization hints, risk awareness feedback

Example: Rejected - Rack power exceeded

Suggested alternative: Rack R14 with 2.3KW free capacity

6 Maintainable Modular Architecture



System designed using clear layer separation:
Config, Monitoring, Derived state, Twin graph, Simulation

7 Reusable and Extensible Design



Architecture supports extension:

- Can add ML prediction
- Cost optimization
- Energy optimization
- Failure prediction
- Auto scaling

Loose coupling + modular services

8 Human-in-loop Safe Automation



Includes approval workflow

- Governance compliance
- Operational safety
- Controlled automation
- Enterprise readiness
- Audit friendly decisions



PERFORMANCE

In-memory derived state



SCALABILITY

Polyglot persistence



MAINTAINABILITY

Layered architecture



REREUSABILITY

Modular services

Our system is not just a Digital Twin. It is a real-time infrastructure decision intelligence platform.

THANK YOU!